

# Generative AI and LLM-based Agents

## Fundamentals of Digitalization and Digital Trends — Lecture 01

Prof. Dr. Michael Granitzer

14 June 2026

# Where we are going

---

**Today's arc:** Machine Learning → Generative AI → Computing with Symbols → Agents  
→ What is left for humans?

# Five steps, one question

**1**

Machine Learning

**2**

Generative AI

**3**

Symbols & Tools

**4**

Agents

**5**

The Human

The technology is only half the lecture. The other half is the question it forces on us.

# 1 — From Machine Learning to Generative AI

---

**Guiding question:** What does it mean for a machine to *learn*, and what is new about *generating* text and images?

# What is machine learning?

A classic, precise definition:

"A computer program is said to **learn** from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in T, as measured by P, improves with experience E." — Mitchell (1997) (Mitchell 1997)

We do **not** write the rules by hand. We specify a task, a way to measure success, and let the program improve from data.

# The main kinds of learning



# Machine Learning

```
graph TD; ML[Machine Learning] --> S[Supervised]; ML --> U[Unsupervised]; ML --> R[Reinforcement]; S --- S_desc[learn from labelled examples (input → label)]; U --- U_desc[find structure in unlabelled data]; R --- R_desc[learn from reward through interaction];
```

## Supervised

learn from labelled examples (input → label)

## Unsupervised

find structure in unlabelled data

## Reinforcement

learn from reward through interaction

**Mitchell (1997): a program learns from experience  $E$  w.r.t. tasks  $T$  and measure  $P$ , if  $P$  on  $T$  improves with  $E$ .**

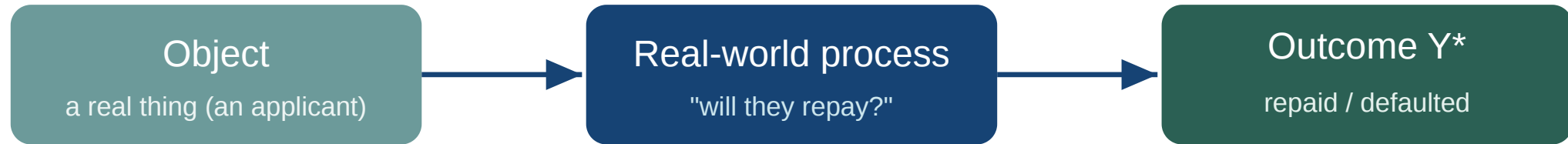
$T$  — task

$P$  — performance

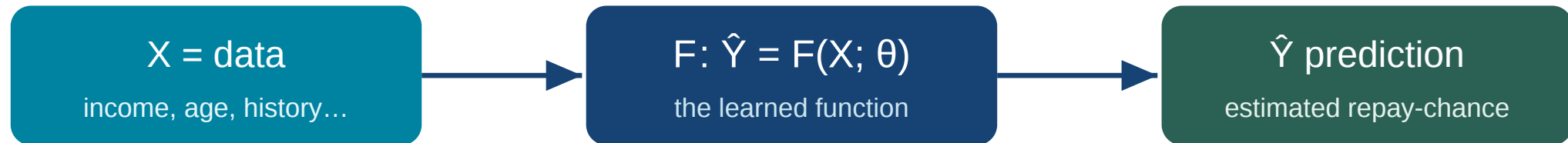
$E$  — experience

# From world to model

REAL WORLD — the process we want to approximate



MODEL — an approximation built from data



*measure / preprocess —  
itself a modelling choice*

*F approximates  
we want  $\hat{Y} \approx Y^*$   
the real process*

Representation gap: reality → model

Both the data X and the function F are *our chosen models* of a messy reality.

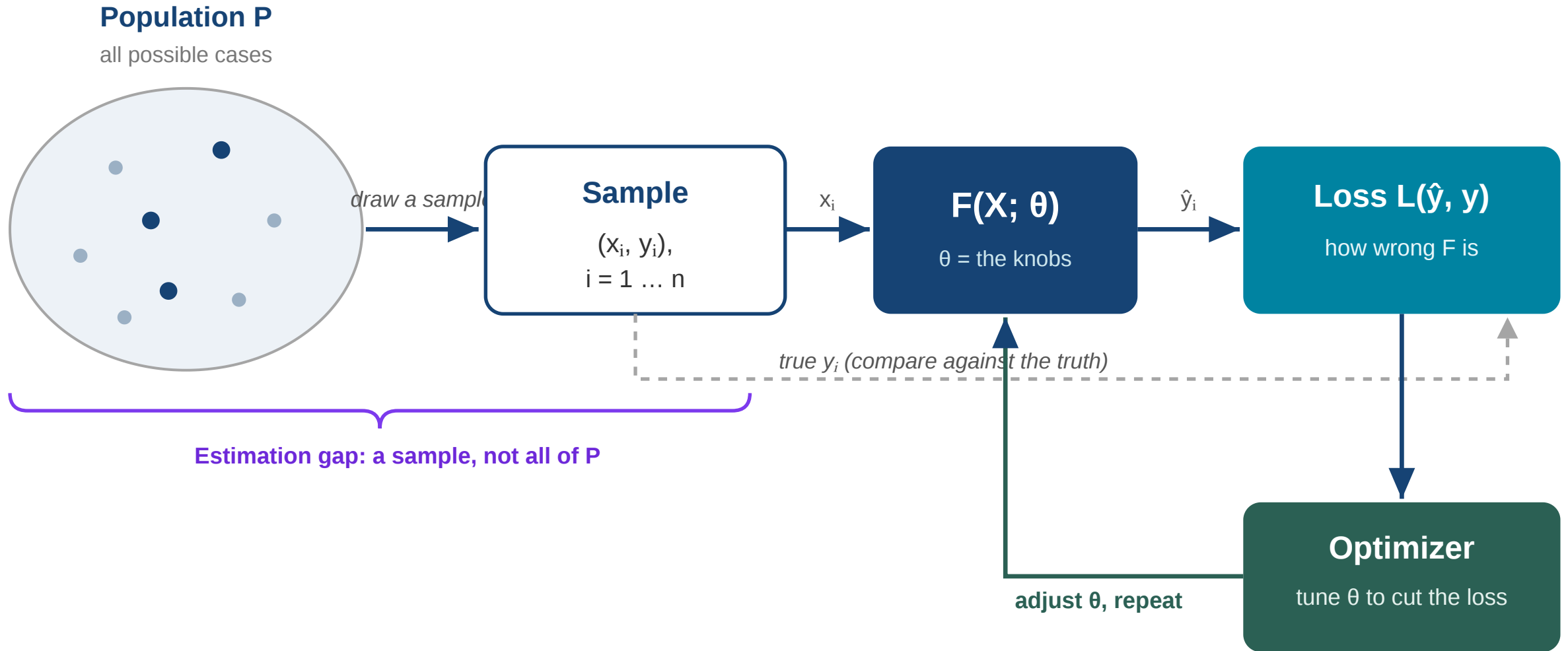




A model is an **approximation of a real-world process**, written  $\hat{Y} = F(X; \theta)$ :

- $X$  — the **data**: our chosen measurement of a real object (*preprocessing is already a modelling decision*)
- $F$  — the **function** standing in for the real process;  $\theta$  are its tunable **parameters** ("knobs")
- $\hat{Y}$  — the **prediction**; we want it close to the true outcome  $Y^*$

# Learning = estimating $\theta$ from a sample



$$\theta^* = \operatorname{argmin}_{\theta} \sum_i L(F(x_i; \theta), y_i) \text{ — pick the knobs that best fit the sample}$$



We never see the whole world — only a **sample**  $(x_i, y_i)$  drawn from a population. Training **tunes the knobs** to fit that sample:

$$\theta^* = \arg \min_{\theta} \sum_i L(F(x_i; \theta), y_i)$$

The **loss**  $L(\hat{y}, y)$  measures *how wrong* the model is; the **optimizer** turns the knobs to reduce it.

Two consequences of the gaps: the model can be **confidently wrong** on cases it never saw, and a **skewed sample** → **a biased model**.

# Predictive learning: the same picture, filled in

Here  $Y$  is a **label** we want to predict — same skeleton, only  $X$  and the label change:

| Task   | X (input)                       | F          | Y (label)       |
|--------|---------------------------------|------------|-----------------|
| Credit | applicant: income, age, history | classifier | repay? yes / no |
| Spam   | the email text                  | classifier | spam / not spam |
| Vision | the pixels of a photo           | classifier | cat / dog       |

The leap to generative AI is just a change in **what  $Y$  is**.

# Discriminative vs. generative

Same skeleton  $\hat{Y} = F(X; \theta)$  — the difference is **what  $Y$  is**:

## Discriminative — $Y$ is a label

A verdict about the input.

- spam / not spam
- credit risk score
- cat vs. dog

## Generative — $Y$ is the raw signal

New content of the same kind as the data.

- write a paragraph
- draw an image
- compose audio

The leap of generative AI: it produces the **raw signal** — text, pixels, sound — not just a verdict about it.

# Generative learning in the same picture

A language model is just our skeleton with a special choice of  $X$  and  $Y$ :

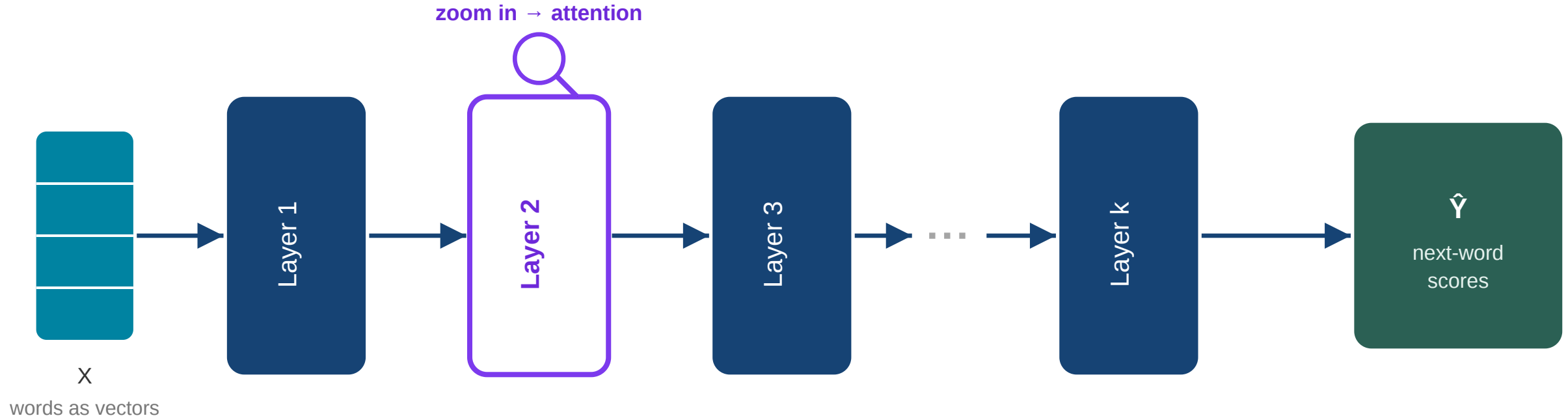
- $X$  = the words so far, ;;  $Y$  = the next word, ;;  $\hat{Y} = F(X; \theta)$
- **self-supervised**: every sentence is its own training data — the next word *is* the label, so no human labelling is needed
- loss = "did it predict the actual next word?"; the optimizer tunes  $\theta$  over **billions** of sentences

Generate by repeating: predict the next word, append it, feed it back in.

So *what is inside  $F$*  that turns "the words so far" into a good next word?

# Inside F: a stack of layers

## Inside F: a stack of layers



$$F = L_k \circ \cdots \circ L_2 \circ L_1$$

*each layer reshapes the representation a little*

$F$  is built from many simple **layers** stacked in sequence —  $F = L_k \circ \cdots \circ L_1$ . Each layer takes the current representation and **reshapes it** a little. Zoom into one layer and you find the

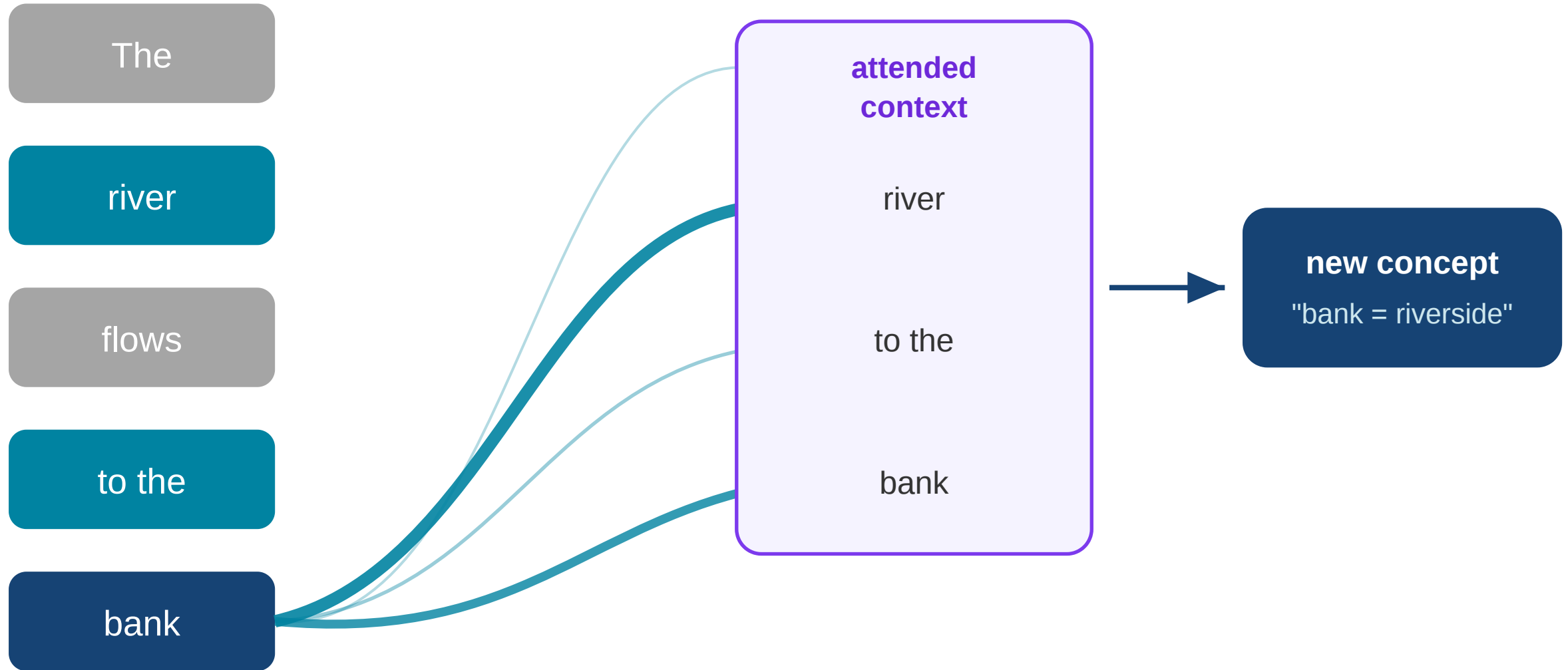
transformer's key idea: **attention**.



# Attention: what one layer does



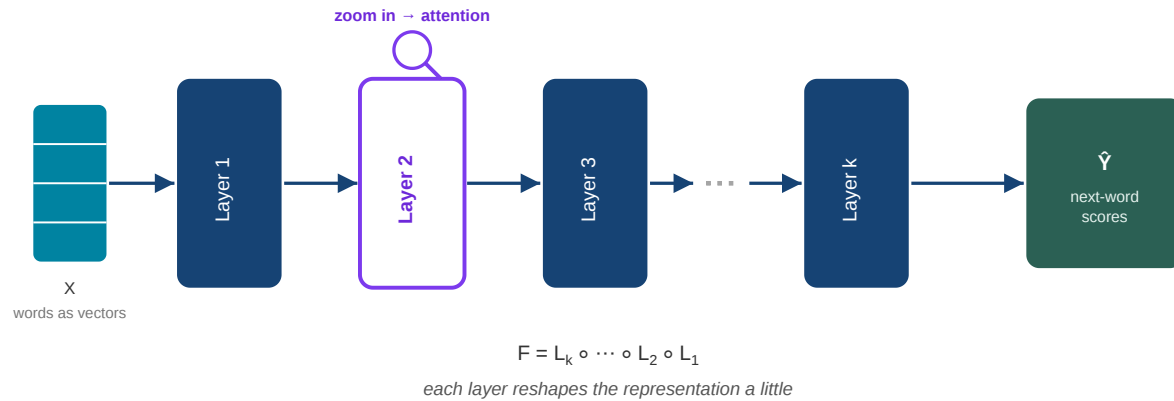
# Attention: relate every word to every other, then build a new concept



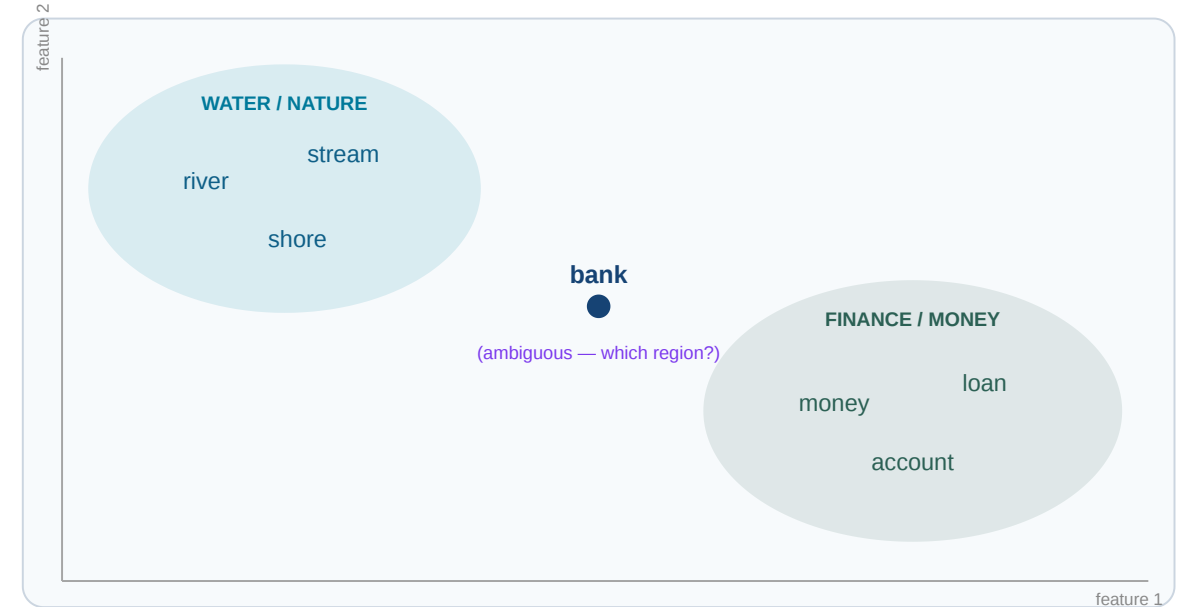
thick = strongly related (attended to)

# Layers move points across a map of meaning

Inside F: a stack of layers



A 2D map of meaning — related words sit close together

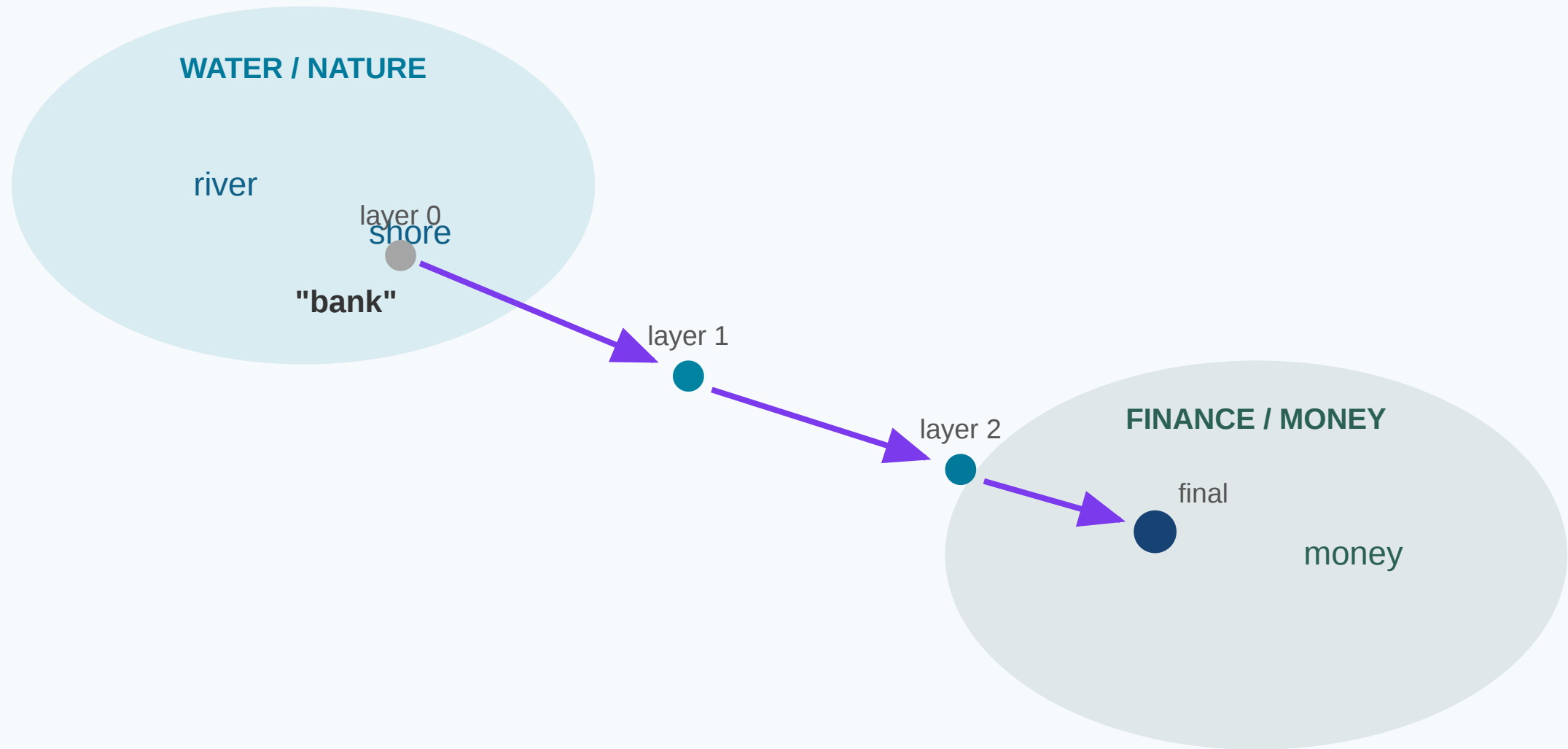


Picture every word as a **point on a map of meaning**, where related words sit close. **One layer = one step** that moves each point across the map.

# Each layer is a step across the map



Each layer moves the word through the map — toward the next word



"deposit money at the ..." pushes "bank" into the finance region → predict **money**

# 2 — Computing with Symbols: a lot, or a parrot?

---

**Guiding question:** Is "predict the next symbol" real reasoning, or a *stochastic parrot* repeating patterns?

# The challenge

A model that predicts the next symbol can also predict the next **digit**, the next **step of a calculation**, the next line of an **argument**.

Is manipulating symbols correctly the *same* as understanding — or is it *enough* to be useful?

# Find the error

A worked long division — **one step is wrong**. Where?

$$\begin{array}{r} 473 \\ 6 \overline{) 2840} \\ \underline{24} \phantom{0} \\ 44 \\ \underline{42} \phantom{0} \\ 20 \\ \times 18 \phantom{0} \end{array} \quad \begin{array}{l} \leftarrow 6 \times 7 = 42 \\ \leftarrow 6 \times 3 = 18, \text{ but } 20 - 18 = 2, \text{ remainder shown as } 0 \end{array}$$

The model wrote the multiplication right but botched the **subtraction / remainder** at the last step. Locally every line *looks* fluent — the error is global.



# Nobody does long division by hand

So what do humans actually do? We **offload** exact symbolic work to a tool.

**Brain**

judgement, goals

**Calculator**

exact arithmetic

**Result**

fast & correct

The skill that matters is not doing every step in your head — it is knowing **what to compute** and **delegating** it to something reliable.

# Symbols are not just digits

Arithmetic is symbolic — but so are **processes**, **procedures**, and **programs**.

## Arithmetic

$473 \times 6 \dots$

## Process

step 1 → step 2 → step 3

## Program

code that runs

The **same next-symbol machine** that predicts digits can predict the next **step of a workflow** or the next **line of code**.

So an LLM can do more than call one tool — it can **write and drive a whole process**: a **controller of a workflow** that invokes many tools in sequence. That is an agent.

# 3 — LLM-based Agents

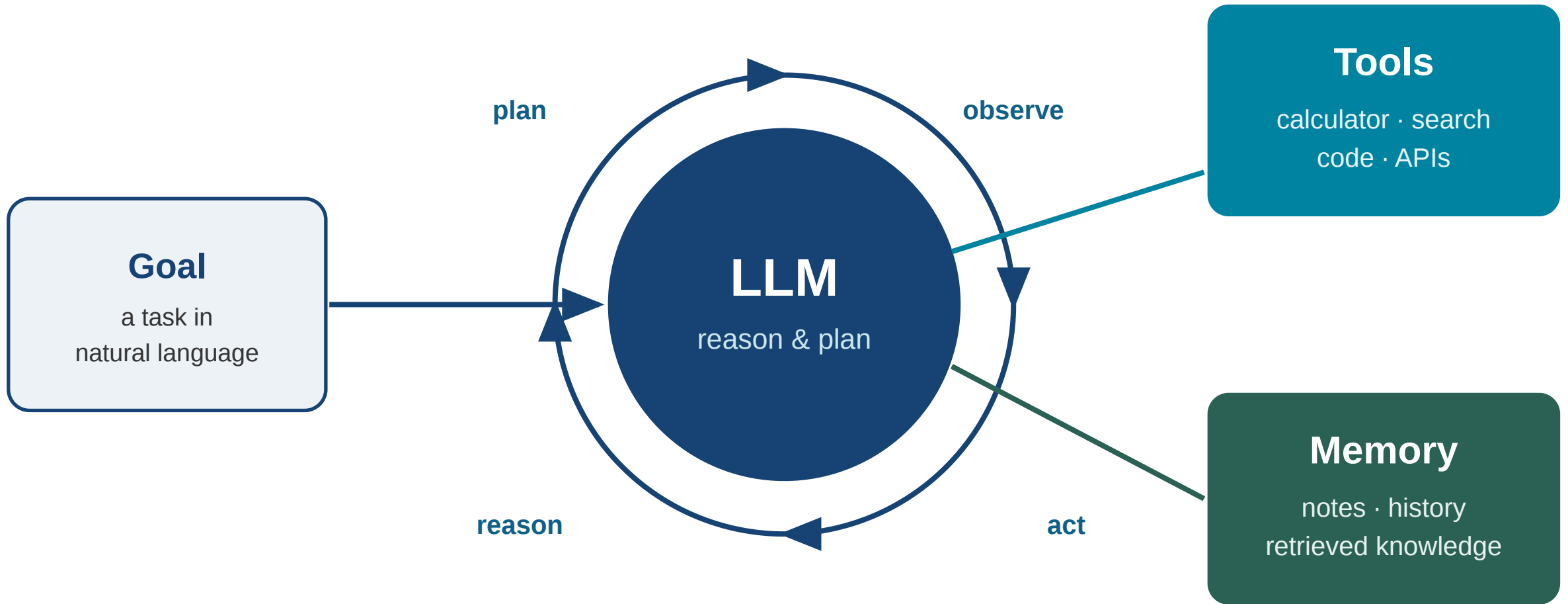
---

**Guiding question:** How does a text predictor become a system that *acts* — searching, computing, remembering?

# Agent = LLM + Tools + Memory

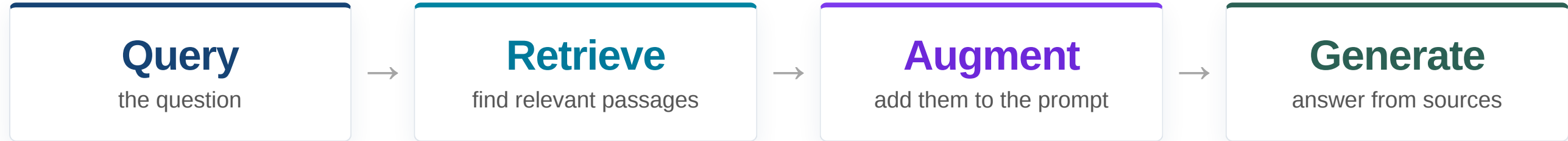


# Agent = LLM + Tools + Memory



# Giving the model knowledge: RAG

**Retrieval-Augmented Generation** — look things up *before* answering.



# From retrieval to agentic search

## Plain RAG — *one shot*

retrieve once → answer.

Good for direct, single-fact questions.

## Agentic search — *a loop*

search → read → refine the query →  
search again → synthesise.

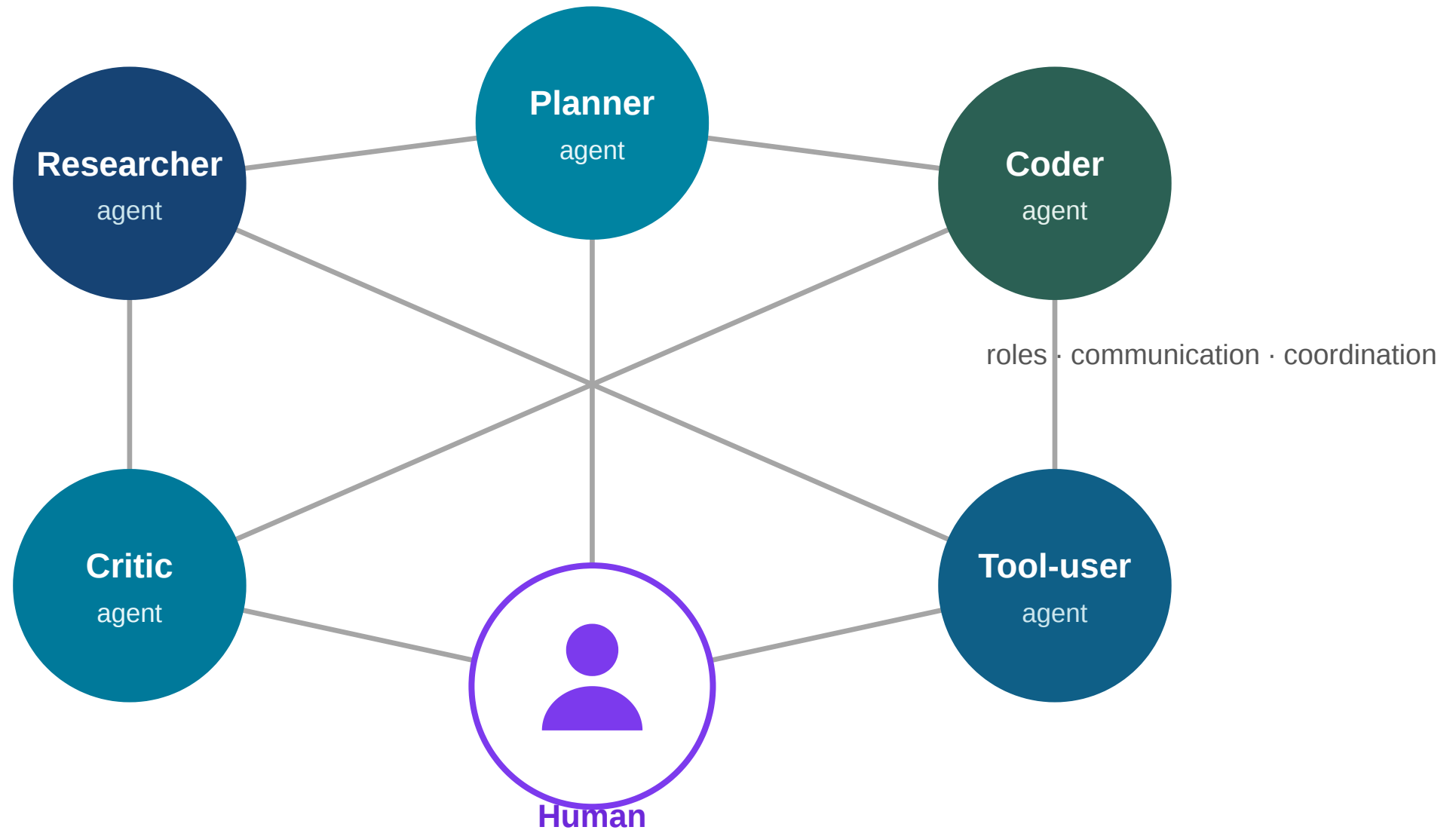
The agent decides *what to look up next*,  
based on what it just learned.

# Many agents, not one





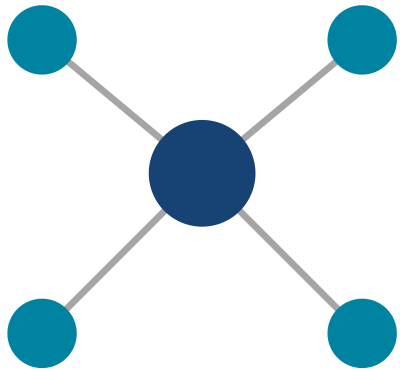
# Not one super-brain — a society of specialised agents and humans



# How agents coordinate

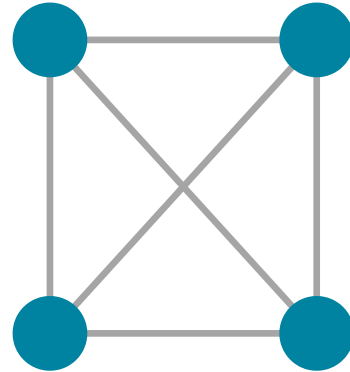
## How the agents are wired together

Centralized



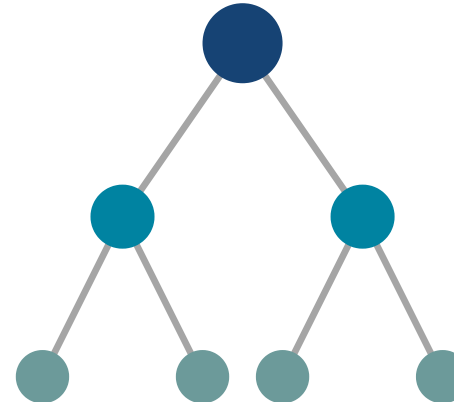
one coordinator

Decentralized



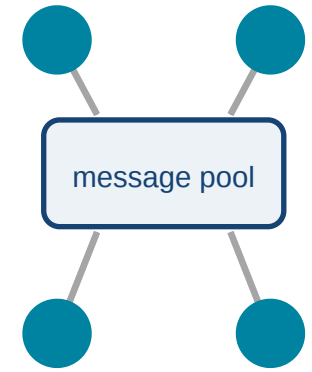
peer to peer

Layered



hierarchy

Shared pool



shared blackboard

Familiar from human organisations: a manager, a flat team, a hierarchy, or a shared workspace.

**Cooperation** — agents build on each other toward a shared goal.

**Debate** — agents argue opposing views; the clash *surfaces errors* and improves the answer.

# Simulating whole worlds

Beyond *solving* tasks, multi-agent LLMs are used to **simulate** them (Guo et al. 2024):

## Societies

opinion & social-media dynamics

## Economies

markets, trading, policy

## Populations

epidemics, mobility

Give agents personas, let them interact, and study the **emergent** behaviour — a new instrument for the social and economic sciences.

Capability comes from **LLM + tools + memory + coordination** — and the same machinery becomes a *laboratory* for studying human systems.

# 4 — Limits and Human–AI Collaboration

---

**Guiding question:** What can AI *not* do for us — and what does that leave for humans?

# A better interface beats raw power

In *freestyle* (centaur) chess, teams of **amateurs with laptops and a good process** beat both grandmasters and supercomputers (Kasparov 2017).

Kasparov's lesson: *weak human + machine + better process > strong human + machine + worse process, and > machine alone.*

The decisive variable was not horsepower — it was the **interface** between human and machine.

# AI does not repeal hard problems

Some problems are hard for *anyone* — human or machine — because of their **computational complexity**.

Even **approximate** Bayesian inference is **NP-hard** (Dagum & Luby 1993). Scaling a model bigger does not dissolve an NP-hard problem.

More compute makes many things faster. It does **not** abolish the wall that complexity theory puts in front of us.

# There is no single super-brain

Look at human intelligence: it is **not one giant brain**. It is a **society** of brains — specialised, cooperating, correcting each other.

**Unlikely:** one monolithic super-intelligence that does everything.

**Likely:** a **society of agents and humans** — specialised, networked, cooperating (Guo et al. 2024).



# ...but that society is hard to build

The same survey is candid about the **open challenges** (Guo et al. 2024) — each a reason humans stay in the loop:

**Errors compound** — one agent's hallucination propagates through the group.

**Collective intelligence** — getting agents to genuinely *learn together* is unsolved.

**Evaluation** — what does "good" even *mean* for a whole society of agents?

**Cost** — many agents = many model calls = a lot of compute.

# So where does that leave the human?

Karpathy's framing: **intelligence has become computable** — you can buy it by the token.

What becomes scarce is **understanding** — the high-level grasp to *guide* intelligence toward the right ends.

The question shifts from "**can we do it?**" to "**what *should* we do?**" — the **goal**, not the skill.

# What to take away — and to build

As intelligence becomes a commodity, your edge is **understanding and direction**: asking the right questions, setting the right goals, and judging the answers.

For an interdisciplinary world: the technical "how" is increasingly automatable. The "why", the "should", and the "what for" are yours — and they decide whether the technology serves us.

# Summary

---

Machine Learning → Generative AI → Symbols & Tools → Agents → a society of minds  
→ the human as the one who *understands and directs*.

# Discussion

- Where in **your** field is the bottleneck already shifting from *execution* to *understanding*?
- Which tasks would you happily delegate to an agent — and which would you never?
- If the future is a *society of agents and humans*, who is **accountable** when it errs?

# Image Credits & References

Dagum, Paul and Luby, Michael (1993). Approximating Probabilistic Inference in Bayesian Belief Networks Is NP-hard. *Artificial Intelligence*.

Guo, Taicheng and Chen, Xiuying and Wang, Yaqi and Chang, Ruidi and Pei, Shichao and Chawla, Nitesh V. and Wiest, Olaf and Zhang, Xiangliang (2024). Large Language Model Based Multi-Agents: A Survey of Progress and Challenges. *arXiv preprint arXiv:2402.01680*. <https://arxiv.org/abs/2402.01680>

Kasparov, Garry (2017). Deep Thinking: Where Machine Intelligence Ends and Human Creativity Begins. *PublicAffairs*.

Mitchell, Tom M. (1997). Machine Learning. *McGraw-Hill*.

Vaswani, Ashish and Shazeer, Noam and Parmar, Niki and Uszkoreit, Jakob and Jones, Llion and Gomez, Aidan N. and Kaiser, Lukasz and Polosukhin, Illia (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems (NeurIPS)*. <https://arxiv.org/abs/1706.03762>